since it might be that the superpartners are so heavy that they are beyond the reach of this machine as well.

Having said this, if in fact the superpartner particles are found, it will most definitely be strong and exciting circumstantial evidence for string theory.

Fractionally Charged Particles

Another experimental signature of string theory, having to do with electric charge, is somewhat less generic than superpartner particles but equally dramatic. The elementary particles of the standard model have a very limited assortment of electric charges: The quarks and antiquarks have electric charges of one-third or two-thirds, and their negatives, while the other particles have electric charges of zero, one, or negative one. Combinations of these particles account for all known matter in the universe. In string theory, however, it is possible for there to be resonant vibrational patterns corresponding to particles of significantly different electric charges. For instance, the electric charge of a particle can take on exotic fractional values such as 1/5, 1/11, 1/13, or 1/53, among a variety of other possibilities. These unusual charges can arise if the curled-up dimensions have a certain geometrical property: Holes with the peculiar property that strings encircling them can disentangle themselves only by wrapping around a specified number of times.18 The details are not particularly important, but it turns out that the number of windings required to get disentangled manifests itself in the allowed patterns of vibration by determining the denominator of the fractional charges.

Some Calabi-Yau shapes have this geometrical property while others do not, and for this reason the possibility of unusual electric-charge fractions is not as generic as the existence of superpartner particles. On the other hand, whereas the prediction of superpartners is not a unique property of string theory, decades of experience have shown that there is no compelling reason for such exotic electric-charge fractions to exist in any point-particle theory. They can be forced into a point-particle theory, but doing so would be as natural as the proverbial bull in a china shop. Their possible emergence from simple geometrical properties that the extra dimensions can have makes these unusual electric charges a natural experimental signature for string theory.